

PRODUCT OVERVIEW

Molitron Odor Abatement System (MOAS)

Commercial-kitchen exhaust odor abatement

MOAS is a wall-mounted system that atomizes Odor Neutralizer Solution through remote misting nozzles in a commercial-kitchen exhaust airstream. Each application is evaluated using the cooking process, exhaust arrangement, available utility connections, and project conditions.



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01 | APPLICATION

Start with the cooking process and exhaust path

Commercial-kitchen exhaust is not the same from one operation to another. Cooking equipment, food and process type, hood grease-removal performance, exhaust airflow, duct routing, and the relationship between the discharge and surrounding properties can all affect the odor concern and the appropriate response.

MOAS should be discussed as part of that complete exhaust picture. The system introduces Odor Neutralizer Solution into the moving exhaust airstream so that it can mix along the available path before discharge.

When to discuss MOAS

- Commercial-kitchen exhaust odor is affecting project planning or facility operations.
- A new restaurant or food-service project is being designed near neighboring properties, occupied outdoor areas, or outdoor-air intakes.
- Cooking equipment, menu, exhaust airflow, or discharge conditions are changing.
- An existing operation needs an evaluation of the available duct path, utilities, and possible equipment location.
- The project team needs to distinguish odor treatment from particulate filtration.

Set the right expectation

MOAS is not presented as a substitute for the kitchen hood, grease-removal devices, exhaust duct, exhaust fan, or other required project systems. It is also not a guarantee that every odor or visible exhaust condition will be eliminated. Product fit, configuration, and results are project-specific.

INDUSTRY CONTEXT [1]

Commercial-cooking effluent can include grease in particulate and vapor forms, smoke particles, and volatile organic compounds that contribute to odor. Grease removal, particulate filtration, and odor treatment therefore address related but not identical portions of an exhaust stream. This general context is not a MOAS specification or compliance statement.

02 | OPERATION

How MOAS operates

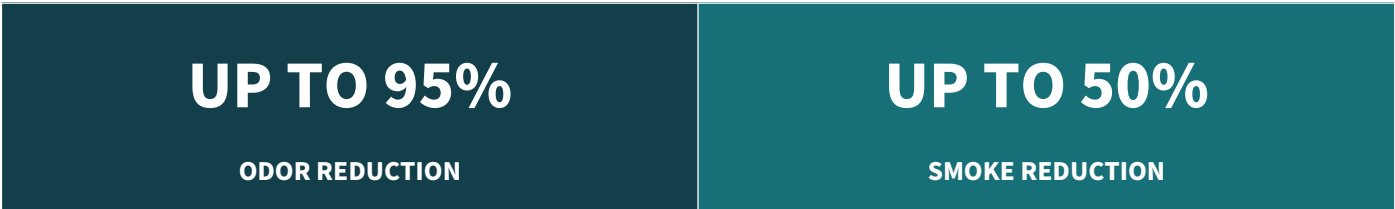
MOAS coordinates the cabinet, Odor Neutralizer Solution, utility connections, remote tubing, and misting nozzles as one operating system.

- 1. **The exhaust fan operates.** An exhaust-fan interlock is intended to prevent MOAS operation when the exhaust fan is off.
- 2. **The cabinet prepares the solution.** Water and Odor Neutralizer Solution are combined within the system for delivery to the exhaust airstream.
- 3. **Air and solution move to the nozzles.** The cabinet's air compressor and solution-delivery components supply the remote nozzle assembly through separate air and solution tubing.
- 4. **The nozzles atomize the solution.** Two remote misting nozzles introduce the solution into the moving exhaust airstream.
- 5. **The exhaust path provides mixing opportunity.** Nozzle location and the available path between atomization and atmospheric discharge are considered during project-specific planning and calibration.

OVERVIEW LIMITATION

This sequence explains operating relationships only. Final installation, wiring, nozzle placement, startup, dilution adjustment, and calibration require controlled project and installation information.

Documentation-derived performance



Important qualification: These documentation-derived figures are upper-limit claims, not guaranteed results for every project. Actual results vary with the cooking process, hood grease-removal performance, exhaust configuration, installation, nozzle location, available mixing/dwell time, solution concentration, calibration, maintenance condition, and other project-specific factors.

Odor treatment and particulate control are different decisions

MOAS is intended for exhaust odor abatement and has documentation-derived smoke-reduction performance. It is not presented as a particulate-filtration replacement. Where a project also requires evaluation of grease or smoke particulate downstream of the hood, Molitron can discuss EPFA separately.

WHY PROJECT INPUTS MATTER [1][2]

Commercial-cooking effluent is a mixture whose particulate and vapor fractions vary among cooking appliances and processes. Collecting accurate equipment and process information supports a useful project discussion; this context does not predict MOAS performance.

03 | SYSTEM OVERVIEW

Follow the complete exhaust path

The MOAS cabinet is located outside the exhaust duct. Separate air and solution tubing connect the cabinet to two remote misting nozzles. The nozzles introduce Odor Neutralizer Solution into the exhaust airstream, and the remaining duct path provides the available mixing opportunity before discharge.

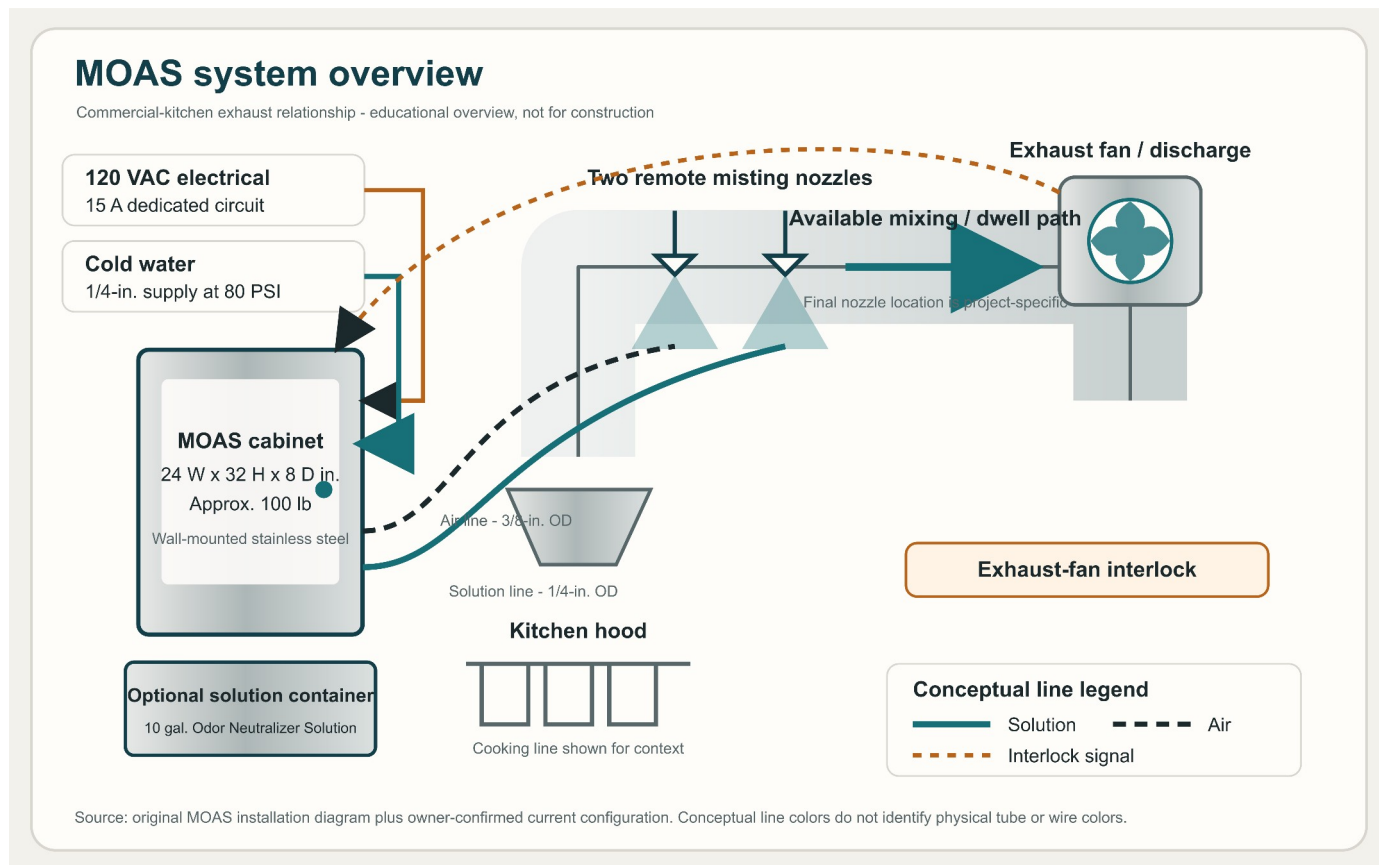


Figure 1. MOAS system overview - educational overview, not for construction.

Nozzle location is project-specific. The final arrangement must account for the exhaust configuration, access, tubing routing, available path to discharge, and controlled installation requirements.

Figure source: original MOAS installation diagram and current owner confirmation. The owner-confirmed electrical value is 120 VAC. Conceptual line colors do not identify physical tube or wire colors.

04 | EQUIPMENT

Components, dimensions, and utilities

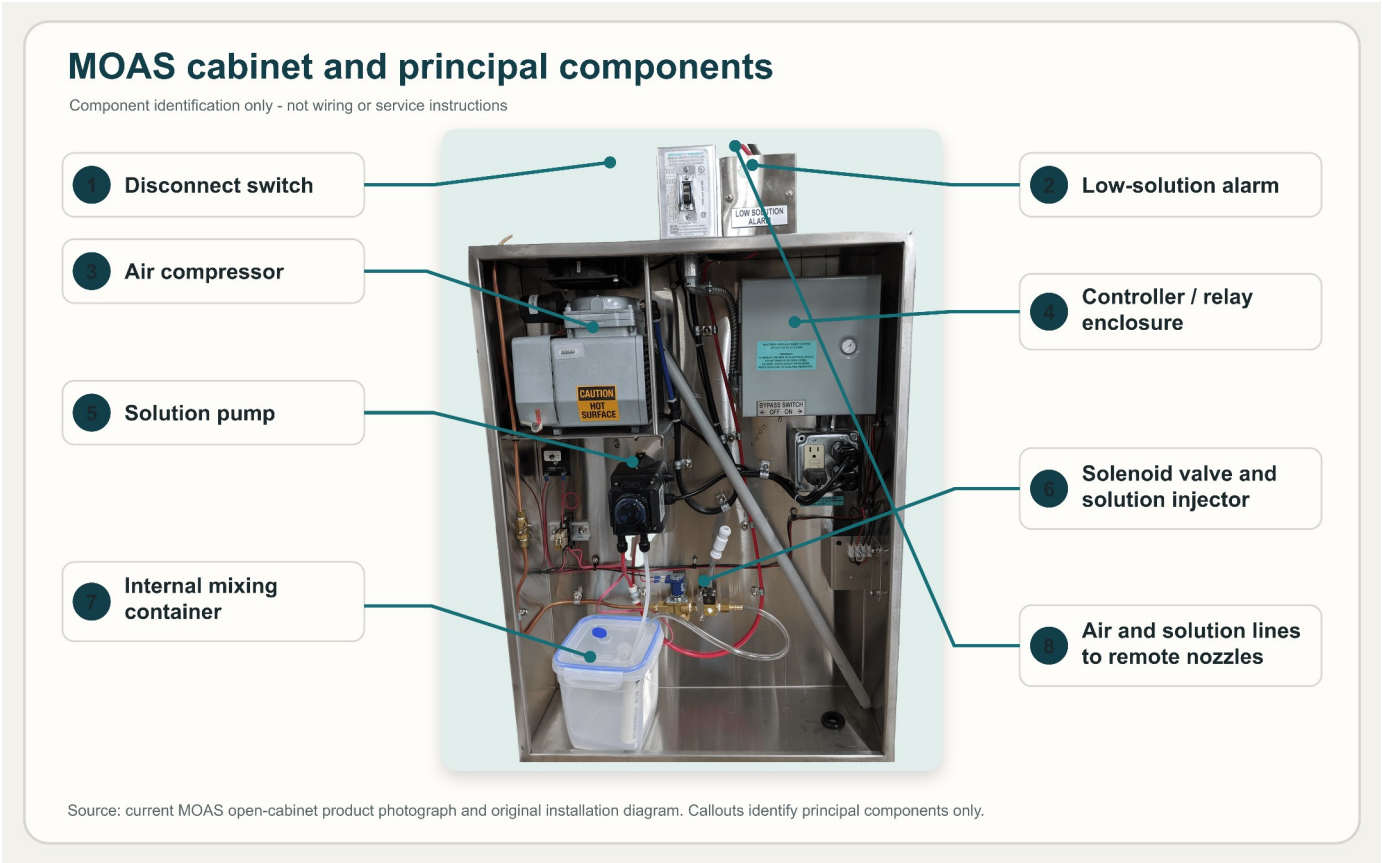


Figure 2. MOAS cabinet and principal components - component identification only.

The wall-mounted stainless-steel cabinet contains the principal operating components. Component callouts are provided for orientation and do not replace wiring or service information.

Assembly	Construction and size	Capacity / approximate weight
MOAS cabinet	18-gauge stainless steel; 24 W x 32 H x 8 D in.	Approx. 100 lb
Optional solution container	16-gauge stainless steel; 24 W x 12 H x 8 D in.	10 gal.; approx. 20 lb empty / 100 lb full

Utility and coordination summary

Requirement	Documentation-derived specification
Electrical	120 VAC, 15 A dedicated circuit
Water	1/4-in. cold-water supply at 80 PSI
Remote tubing	1/4-in. OD solution tubing; 3/8-in. OD air tubing
Operating coordination	Exhaust-fan interlock

Two-nozzle system capacity: Up to 6,000 CFM per two-nozzle system. Final configuration, nozzle location, tubing routing, startup, and calibration are project-specific.

05 | PROJECT PLANNING

Plan the project conversation

The most useful MOAS discussion begins with the actual cooking operation and exhaust system. Available drawings, photographs, equipment schedules, and operating information can help Molitron understand the project conditions.

Information to gather

- Cooking equipment and fuel type
- Menu, food, or process producing the odor concern
- Hood type and grease-removal arrangement
- Exhaust airflow in cubic feet per minute
- Duct routing and approximate available distance from the proposed nozzle location to discharge
- Exhaust-fan and discharge location
- Nearby property uses, occupied areas, and outdoor-air intakes relevant to the concern
- Available 120 VAC electrical power and cold water
- Potential cabinet and optional solution-container location
- Project location, jurisdiction, status, and schedule

Project responsibilities

Role	Project responsibility
Owner / operator	Provides operating information, identifies the concern, maintains access to current instructions, and reports observed operating or service issues.
Engineer / designer	Coordinates MOAS with the hood, exhaust, utilities, controls, access, and other project requirements.
Installer	Installs the equipment using controlled installation information and approved project documents.
Service provider	Performs authorized startup, calibration, and service activities using current technical information.
AHJ	Independently reviews and determines project acceptance under requirements applicable to that installation.
Molitron	Provides product information and discusses the proposed application using the project information available.

OPERATING AWARENESS

MOAS performance depends on the condition and operation of the overall system. Operators should use current Molitron instructions for solution replenishment, alarm response, inspection, and service. This brochure does not establish maintenance intervals or replace controlled operation and service information.

06 | PRODUCT CONTEXT

MOAS and particulate filtration

MOAS and EPFA address different project needs. They may be discussed separately or together when an exhaust project includes both odor and particulate concerns.

	MOAS	EPFA
Primary role	Exhaust odor abatement	Dry particulate filtration
General method	Atomizes Odor Neutralizer Solution through remote nozzles	Uses multiple stages of dry filtration
Project relationship	Can be evaluated as a standalone odor-abatement system	Can be evaluated where downstream particulate filtration is needed

This comparison does not promise combined performance or establish that both products are required. Product selection and system design remain project-specific.



Typical equipment relationship shown in an existing installation. Project conditions vary.

Selected installations documented by Molitron

Chick-fil-A, Chipotle, Burger King, Lazy Dog, District Taco, Farmer's Table, Ritz Carlton, Alexander's Steak House, Luna Grill, Little Italy, Cooper's Pit Bar-B-Que, and Mike's BBQ.

REFERENCE LIMITATION

These names identify selected installations recorded in Molitron's legacy brochure. They do not imply sponsorship, endorsement, partnership, current configuration, or a guaranteed project outcome.

07 | NEXT STEP

Discuss the application with Molitron

Bring the exhaust system into focus

MOAS selection begins with the cooking process, exhaust airflow, hood grease-removal arrangement, duct path, discharge conditions, available utilities, and project goals. Send the information available for your project, and Molitron can help identify the next useful questions.

MOLITRON

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Current ETL Listing

THIRD-PARTY LISTING

MOAS is ETL Listed for the United States and Canada under Intertek Report 101453585DEN-002. The report identifies UL 197 and CSA C22.2 No. 109 as the applicable standards. The listed equipment rating is 120 V, 3.5 A, 60 Hz; project planning separately requires a 120 VAC, 15 A dedicated circuit. Listing does not establish project approval or AHJ acceptance.

Important project limitation

PROJECT-SPECIFIC INFORMATION GOVERNS

MOAS performance is application-specific. Published 'up to' values are not guarantees. Final equipment selection, installation, wiring, nozzle placement, startup, dilution adjustment, calibration, maintenance, and project acceptance must be addressed using current controlled information and requirements applicable to the specific project. This brochure is an educational product overview, not a construction drawing, installation manual, wiring diagram, service manual, certification record, or project approval.

Research-context sources

- [1] [ASHRAE, 2023 ASHRAE Handbook - HVAC Applications, Chapter 34, Kitchen Ventilation](#). Accessed July 13, 2026.
- [2] [U.S. EPA, 2020 National Emissions Inventory Technical Support Document: Commercial Cooking](#). Published March 2023; accessed July 13, 2026.

These sources provide general education only. They do not establish MOAS performance, listing, code compliance, safety compliance, or project suitability.

Document control

Document	Revision	Date
MOAS-BRO-001	Rev B	July 2026

Project questions: [email Molitron](mailto:cleanair@molitron.com)